Artificial Intelligence: Homework 1 Written

Mu RuoTong,225040014

Introduction

INTRO1: Meaning of "Rationally" In AI, acting "rationally" means choosing actions that are expected to maximize a specific utility function or performance measure, given the agent's current knowledge and available information.

- **Example 1:** A self-driving car choosing to brake rather than swerve into traffic, maximizing the utility of passenger safety.
- **Example 2:** Rational investing — Making investment decisions based on fundamental financial data rather than market hype or emotions.

INTRO2: Tasks humans do poorly Humans do not perform well at tasks requiring the rapid, continuous processing of massive datasets. An example is scanning millions of real-time credit card transactions globally to detect subtle patterns of financial fraud.

Search Algorithms

BFS1: Breadth-First Search States Assuming a constant branching factor b, the number of generated states up to level 5, where half of the states on level 5 are searched, is mathematically represented as:

$$1 + b + b^2 + b^3 + b^4 + \frac{b^5}{2}$$

Based on the exact 8-puzzle diagram provided, the goal node is explicitly labeled as node 27. Therefore, 27 states were searched to reach the goal.

DFS1: DFS Search Tree Based on the fractional entry and exit times provided in the graph diagram ($u : 1/12, v : 2/11, w : 7/8, s : 4/5, t : 3/10, r : 6/9$), the DFS traversal perfectly matches the sequence: $u \to v \to t \to s \to$ (backtrack) $\to r \to w$. The tree edges are $(u, v), (v, t), (t, s), (t, r),$ and (r, w).

DFS2: Entering and Leave Times Tracing alphabetical DFS starting from S, the discovery/finish times are:

- $S : 1/16$
- $A : 2/15$
- $B : 3/14$
- $C : 4/5$

- $E : 6/13$
- $D : 7/8$
- $F : 9/12$
- $G : 10/11$

DFS3: Directed Graph Alphabetical DFS Starting at q, the traversal sequence and backtracking is as follows:

1. Start at q, visit s, visit v, visit w.
2. Backtrack to q, visit t, visit x, visit z.
3. Backtrack to t, visit y.
4. Backtrack to root. Pick next unvisited: r.
5. Start at r, visit u. All nodes visited.

Pathfinding

Dijkstra1 Running Dijkstra's algorithm from node a, nodes are added to the visited set S in the following order: 1. $S = \{a\}$ 2. $S = \{a, b\}, b = 4$ 3. $S = \{a, b, h\}, h = 5$ 4. $S = \{a, b, h, g\}, g = 6$ 5. $S = \{a, b, h, g, f\}, f = 8$ 6. $S = \{a, b, h, g, f, c\}, c = 12$ 7. $S = \{a, b, h, g, f, c, i\}, i = 12$ 8. $S = \{a, b, h, g, f, c, i, e\}, e = 18$ 9. $S = \{a, b, h, g, f, c, i, e, d\}, d = 19$

A* 1: Standard Heuristic

- **Iter 1:** Pop $A(g = 0, f = 30)$. Open: $X(f = 44)$.
- **Iter 2:** Pop $X(44)$. Open: $C(f = 45), B(f = 49)$.
- **Iter 3:** Pop $C(45)$. Open: $W(f = 48), B(f = 49), L(f = 81)$.
- **Iter 4:** Pop $W(48)$. Open: $T(f = 48), B(f = 49), K(f = 68), L(f = 81)$.
- **Iter 5:** Pop $T(48)$. Goal reached!

Optimal Path: $A \to X \to C \to W \to T$ (Total Cost = 48).

A* 2: Modified Heuristic ($h(B) = 6$)

- **Iter 1:** Pop $A(0, 30)$. Open: $X(44)$.
- **Iter 2:** Pop $X(44)$. Open: $B(f = 39), C(f = 45)$.
- **Iter 3:** Pop $B(39)$. Open: $C(45), T(f = 50), W(f = 54), S(f = 57)$.
- **Iter 4:** Pop $C(45)$. Updates W. Open: $W(f = 48), T(50), S(57), L(81)$.
- **Iter 5:** Pop $W(48)$. Updates T. Open: $T(f = 48), S(57), K(68), L(81)$.

- **Iter 6:** Pop $T(48)$. Goal!

Conclusion: Yes, the same optimal path is found.
Problem: Reducing $h(B)$ to 6 makes the heuristic inconsistent (non-monotonic) because $h(X) \not\leq c(X, B)+h(B)$ ($27 \not\leq 16+6$). Inconsistent heuristics can cause A* to expand nodes with suboptimal g-values initially, which can require reopening closed nodes to find the true shortest path, harming performance.

AStar3: Polygonal Obstacles

(a) **How many states?** If we use a visibility graph, the continuous coordinate space is reduced to a finite number of states: the start node, the goal node, and the vertices of all polygons.

(b) **Why along the edge?** In Euclidean geometry, the shortest path between two points is a straight line. A path will only deviate from a straight line when obstructed by an object. Thus, the shortest path must stretch tightly between the obstacle corners (vertices), resulting in straight line segments.